Experiment on Charging a Capacitor

Learning Objectives

- Analyze charging voltage variation over time.
- Calculate the time constant (τ).
- Experimentally measure capacitance (C).
- Calculate percentage error in capacitance measurement.

AIM OF THE EXPERIMENT

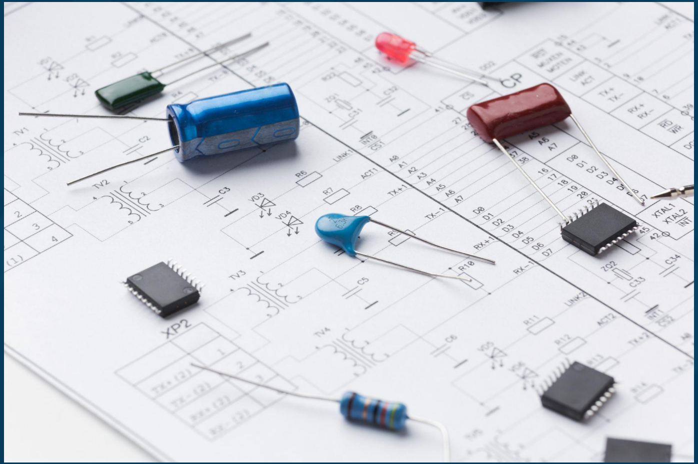
To investigate the charging characteristics of a capacitor and determine its capacitance and time constant.

Apparatus for the Experiment

- Voltmeter.
- Power supply.
- Charging circuit (Capacitor, Resistor, Switch key).
- Stopwatch.
- Wires.
- Praxilabs virtual laboratory – RC Circuit Charging

THEORY: what is a Capacitor

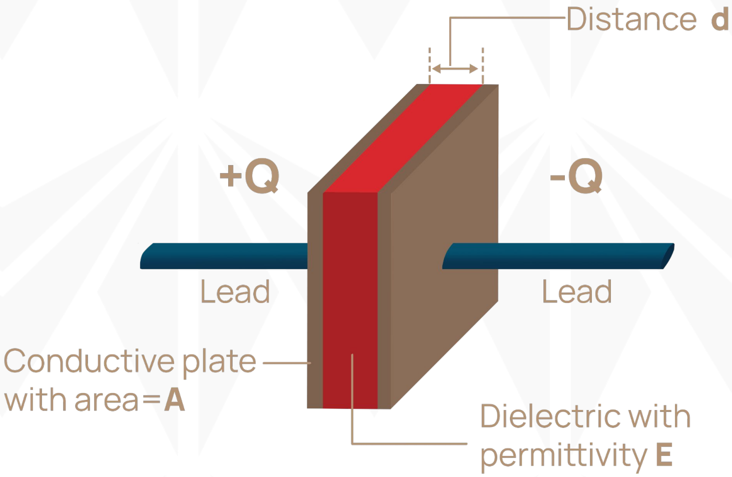
- A capacitor is a device that stores electric energy in an electric field form.
- Capacitors have several uses, such as filters in DC power supplies and as energy storage banks for pulsed lasers.
- Capacitors are electronic components that can pass alternating current (AC) while blocking direct current (DC).



THEORY: WHAT IS A CAPACITOR

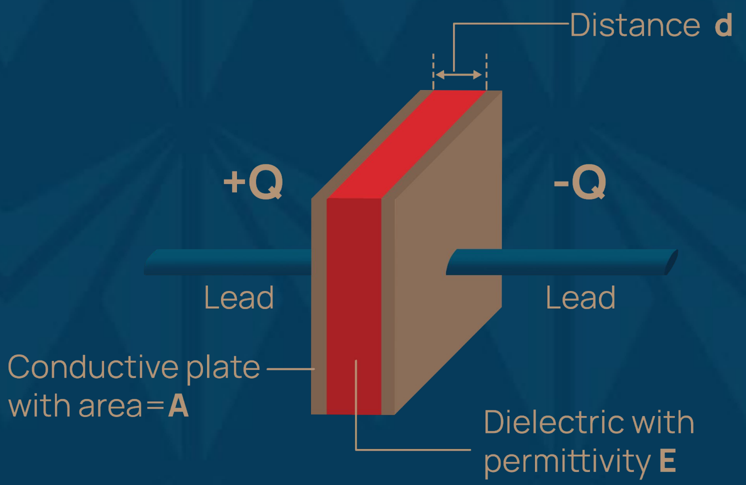
A capacitor consists of two conductors separated by a small distance, d, plate Area, A and dielectric material with permittivity, ε.

$C = \varepsilon A / d$



THEORY

- When the conductors are connected to a charging device (for example, a battery),
- The charge is transferred from one conductor to the other until the difference in potential between the conductors becomes equal to the potential difference between the terminals of the charging device.



Charge on a capacitor

- The amount of the charge stored on either conductor is directly proportional to the voltage, and the constant of proportionality is known as the capacitance.

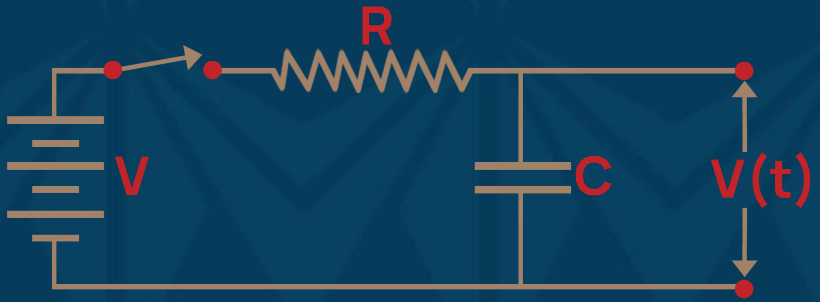
  $Q = CV \quad (1)$
- The charge Q is measured in units of coulomb (C),
- the voltage V in volts (V), and
- The capacitance C in units of farads (F).
- The capacitors are physical devices; capacitance is a property of devices.

RC CIRCUIT

- In the RC circuit shown:
- Turn on the power supply and turn on the switch key.
- The capacitor starts charging with time from zero until reaching its maximum voltage (V_0) according to equation (2) below.

$$V(t) = V_0 (1 - e^{-t/\tau}) \quad (2)$$

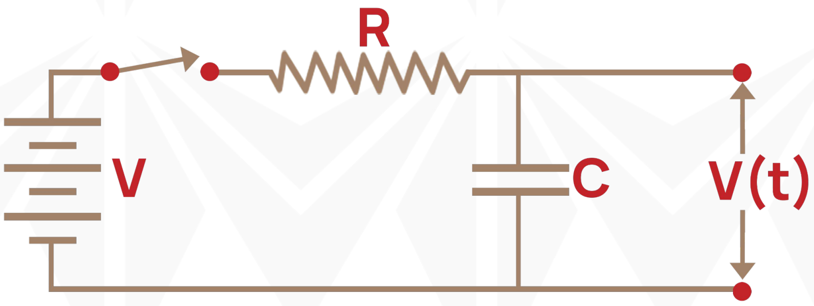
where τ is the time constant, V(t) is voltage on capacitor during charging at time t



THEORY

- The time constant (τ) of a circuit component, such as a resistor-capacitor (RC) circuit, is a measure of how quickly the voltage or current across the component changes in response to a sudden change in input.
- The time constant depends on the resistance (R) and capacitance (C) values in the circuit and is calculated using the following equation:
- $\tau = RC$ (3)

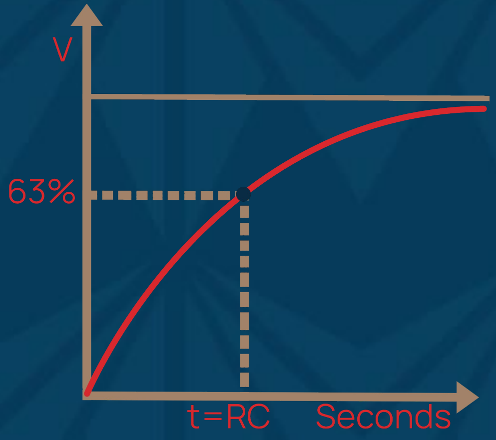
- τ (tau) is the time constant measured in seconds (s).
- R is the resistance in ohms (Ω).
- C is the capacitance in farads (F).



τ

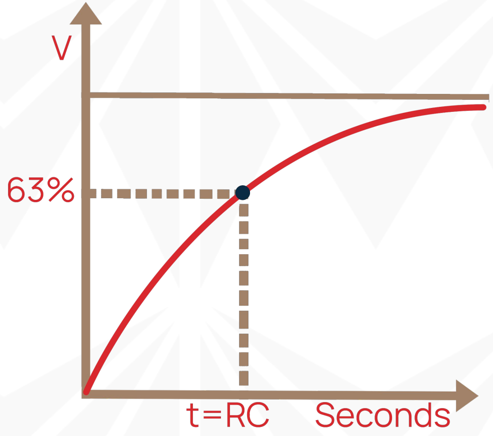
THEORY

- The time constant represents the time it takes for the voltage across the capacitor in an RC circuit to charge to approximately 63.2% of its maximum value (charging) or discharge to approximately 36.8% of its initial value (discharging) in response to a step change in voltage or current.



THEORY

The time constant is a fundamental parameter used to describe the behaviour of RC circuits in terms of their time-dependent responses, such as charging and discharging processes.



MID-LESSON QUESTIONS

Q1: What is a capacitor?

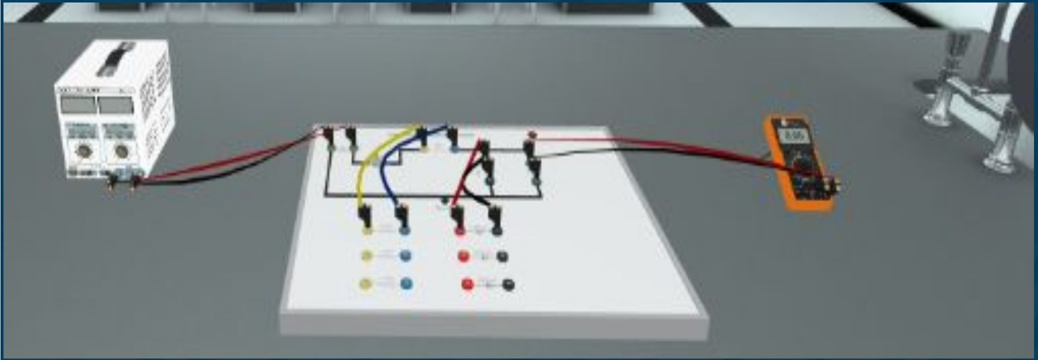
Q2: In an RC circuit, if the resistance (R) is increased while keeping the capacitance (C) constant, what happens to the time constant (τ) and the charging time (t_charge)? Explain.

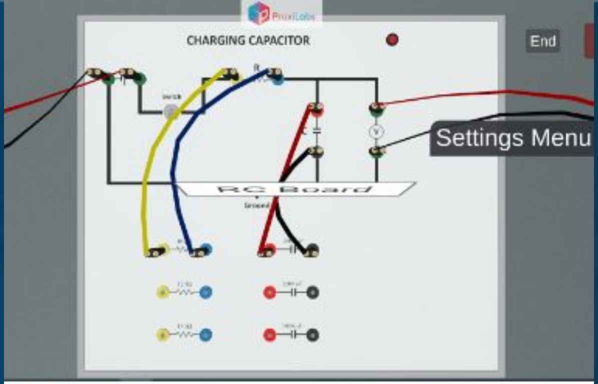
MID-LESSON ANSWERS

A1: A capacitor is a device that stores electrical energy in an electric field. It is made up of two conductors, called plates, separated by an insulator.

A2: Answer 1: When the resistance (R) in an RC circuit is increased while keeping the capacitance (C) constant, the time constant ($\tau = R \times C$) also increases. As a result, the charging time (t_{charge}) also increases. This means that it takes longer for the voltage across the capacitor to reach a specific fraction of the source voltage during the charging process. In other words, the circuit charges more slowly with higher resistance.

PROCEDURE



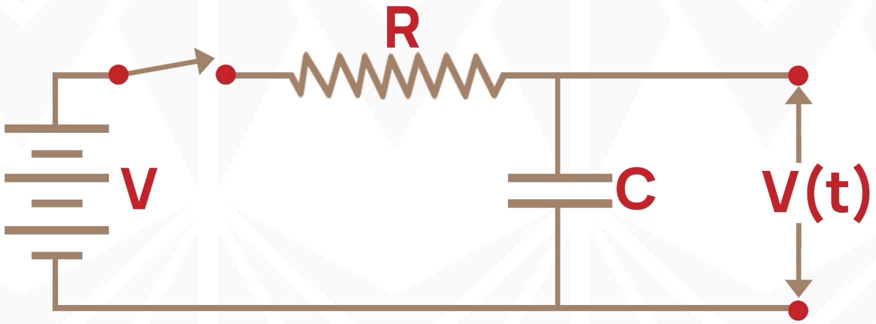


1. Connect the power supply and voltmeter in the charge circuit then choose a suitable capacitor and suitable resistor to connect them by wires in the charging circuit as shown.

PROCEDURE

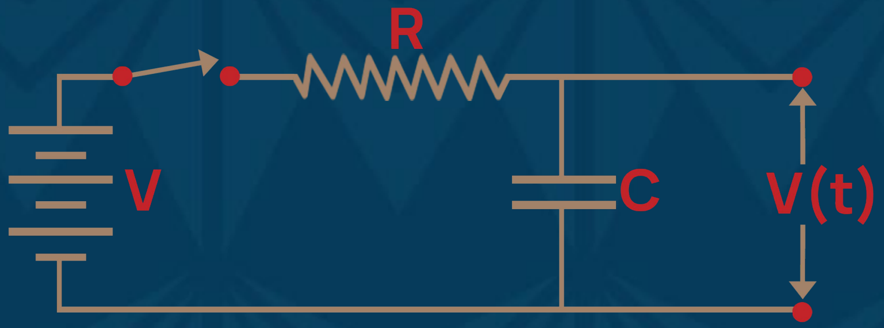
2. Turn on the power supply and choose the output voltage, for example, 15 volts.
3. Turn ON the switch key and with a stopwatch record the value of the voltmeter every 10 seconds until maximum volt.

Note: take 10 measurements at least.



PROCEDURE

4. You may end the experiment after any trial by clicking the “End” button.
5. Once you click the “End” button, an excel sheet will be downloaded containing the recorded data.



PROCEDURE

6. Data download should look like this:

Trial #1:

V_0 (Volt)	1
R (kiloOhms)	10
C (microfarads)	1000

t(sec)	V(Volt)
5.7	0.43
9.57	0.62
13.27	0.73
16.57	0.81
19.32	0.86
22.19	0.89
24.84	0.92
28.64	0.94
34.07	0.97
37.59	0.98

Data Analysis

1. Plot the relation between the voltage on the capacitor (on the y-axis) and time (on the x-axis).

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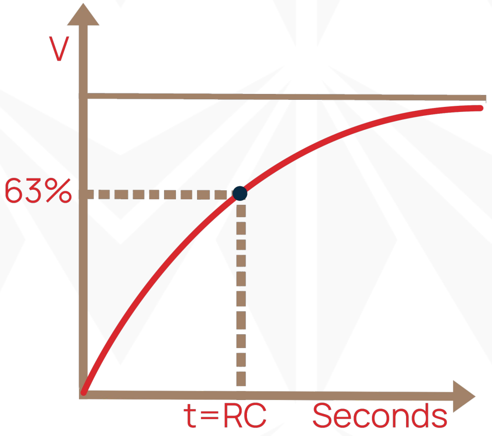
Data Analysis

2. From the curve, determine the experimental value of the time constant, τ. (Determine the 63.2% of the maximum voltage and trace the in the graph to the time axis as shown in the figure)



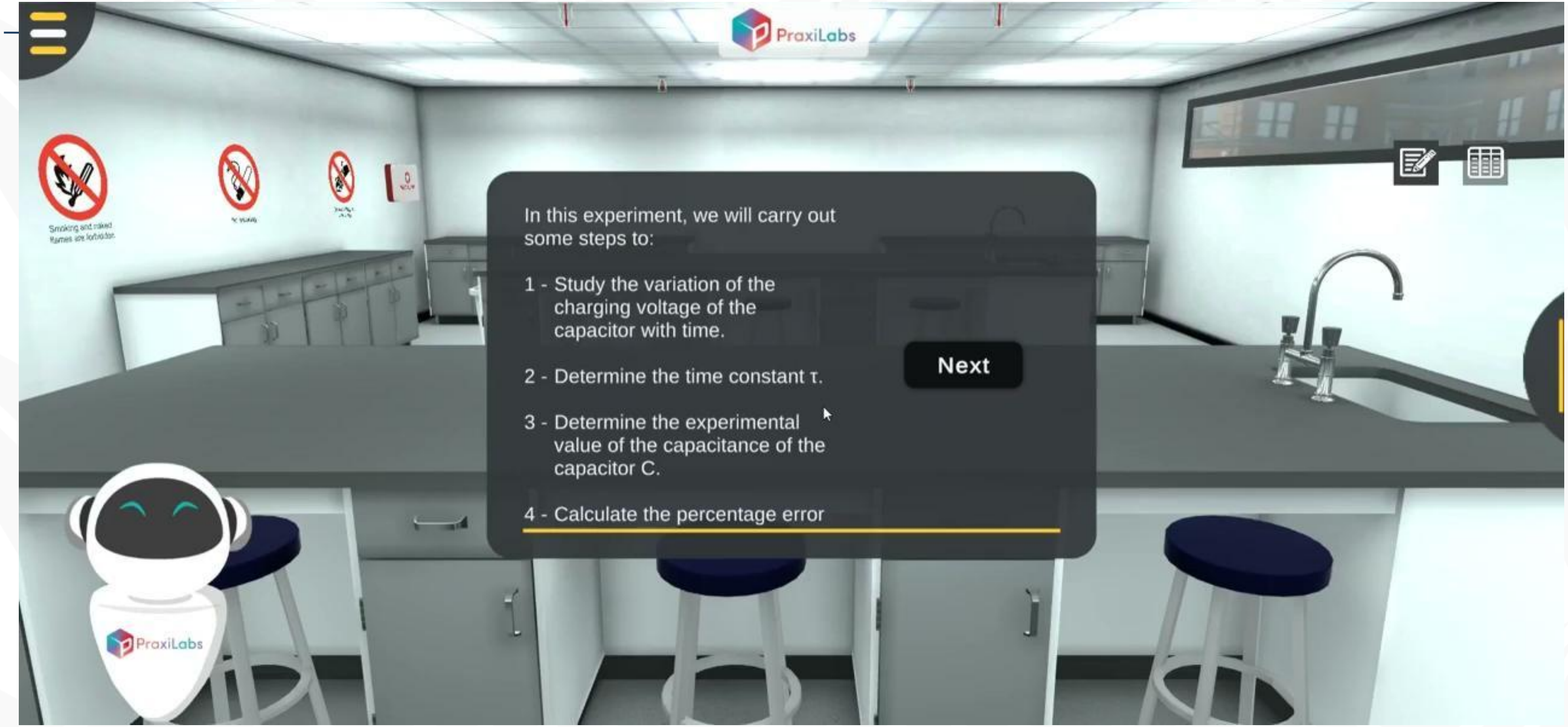
$$V(t) = V_0(1 - e^{-1}) = 0.632\, V_0$$

DATA ANALYSIS

3. Given the value of resistance, you can get the value of capacitance according to the following equation: $C = \tau / R$
4. Calculate the percentage error of the capacitance using the following equation.

$$\%\ \text{error} = \left| \frac{\text{Theoretical value} - \text{Experimental value}}{\text{Theoretical value}} \right| \times 100$$

WALK_THROUGH VIDEO



FURTHER READING

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Serway, R.A. and Jewett. J.W. (2014). Physics for Scientists and Engineers with Modern Physics, Ninth Edition. Physical Sciences: Mary Finch.

FURTHER READING

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Young, H. D. (2012). Sears & Zemansky's college physics. 9th ed. Pearson Education, Inc.

http://hyperphysics.phy-astr.gsu.edu

http://www.physicstutorials.org

Thank You